



BAI

DevOps Foundations Continuity and LMS Microservice Design/Architecture Statement of Work

Prepared For: Jeff Dunn
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Executive Overview

nvisia is collaborating with BAI to execute on an architectural roadmap to support the product vision for the Learning & Development (L&D) platform. BAI converted its customers from Saba to the current Latitude-based learning management system (LMS) in 2017. The LMS has since served BAI's needs but does not effectively provide a foundation for BAI's product vision. nvisia is partnering with BAI on their strategic objective to deliver a modern software product aligned with the proposed application architecture. This will be a phased approach over the next few years.

It's to be noted that the project team faced challenges with containerization during Phase 1. (These details are outlined in the LMS DevOps Situation Analysis document drafted by BAI). Nvisia successfully containerized six LMS components in Phase 1. However, due to complexities with upgrading certain components to .NET Core and a growing backlog of technical debt, major LMS components (including LMS and Latitude) were not addressed because of the high estimated effort required would far exceed the original budget and timeline.

This Statement of Work (SOW) will continue cloud DevOps foundational work for the prioritized LMS components; continue to move each containerized LMS component through local, Dev, Staging and Production environments to prove the working solution; create and test new environments; continue to provide support/training to BAI staff to leverage the new DevOps environments, tools, and practices; implement a new logging/monitoring solution and performance dashboard configuration; collaborate with BAI to design an LMS microservices architecture and build a single proof of concept (POC) microservice; analyze and create a backlog for addressing chronic database issues affecting the LMS; and create a SSO/Authentication strategy and implement solution for a single application.

Deliverables

The deliverables below have been identified as key deliverables for this project. They are listed in the expected order of completion. This list may change as nvisia and BAI begin working on the engagement and adapting as the project progresses.

Project Management Artifacts (to be delivered by nvisia):

1. Phase 2 Project Plan and Project Charter
2. Weekly status updates - nvisia consultants will provide a weekly status update to BAI staff
3. Documentation
 - a. nvisia will store all documents and deliverables produced for this project in an electronic document library supplied by BAI so that information can be versioned.
 - b. Stored documents will include (but not limited to): project plan, resource plan, risk register, product documentation for 3rd party tools, training materials, installation and configuration notes, project status reports, decision log.
 - c. During the project wrap-up, NVISIA will conduct knowledge transfer sessions to walkthrough the documents with BAI staff. nvisia to update documentation based on feedback.

Project Approach and Deliverables by Category:

Throughout the engagement, there will be quality assurance through early review of plans with clear outcomes aligned to business objectives. In addition, proper overviews and artifact reviews will be conducted toward the end of each activity to ensure plan goals are being met, or to understand re-alignment decisions.

Knowledge transfer will be evidenced by completion of training plan detailed in the project charter, which may include activities such as self-study, working sessions, overviews, Lunch and Learns, and hands-on work. Lunch and Learn events would ideally have a BAI resource leading to further push thought ownership. The



success criteria is BAI's ability to take ownership of deliverables and able to interact independently without nvisia's support.

Item	Activities	Deliverables	Done Criteria	Purpose	Notes
1a. Monitoring <i>2-4 weeks</i>	▪ Implement 3rd-party monitoring solution ▪ Install at system, front-end, and back-end levels ▪ Configure performance dashboard	▪ Configured performance dashboard ▪ Insights into critical/chronic LMS performance issues	BAI resource verified setup for APM Dashboard access, alerts	Significantly increase ability to identify specific sources of LMS issues and inform priorities	Requires BAI to purchase monitoring tool (\$18k/year)
1b. Microservices <i>3 months (includes database considerations)</i>	▪ Design microservice architecture ▪ Build microservice POC	▪ Plan and documentation for as is/to be microservice architecture ▪ Microservice POC	Business deliverable documentation reviewed with BAI, POC running in Dev, and ownership transferred to designated resources	Foundational work required to move to a microservices architecture and demonstrate value	Requires BAI to hire Solutions Architect @100% allocation (with help from nvisia)
1c. Database <i>1 month (overlap with microservices analysis)</i>	▪ Address major performance issues with LMS database ▪ Advise on version upgrade or migration to Azure SQL	▪ Backlog for database migration ▪ Supporting documentation	Business deliverable documentation and backlog reviewed with BAI and ownership transferred to designated resources	Address chronic LMS DB performance issues that generate most outage calls	Monitoring and microservices analysis will inform this work
2. Authentication <i>4 months</i>	▪ Implement Azure B2C SSO Authentication ▪ Build POC for current/future architecture	▪ One application using SSO in Dev ▪ Migration roadmap for BAI apps to SSO/B2C	Identify BAI owners; ensure owners have access credentials to execute flow for verification	Build scalable, flexible system authentication and single sign on (SSO) solution that supports	Requires BAI to hire Solutions Architect @100% allocation (with help from nvisia)

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			SSO working in Dev	multiple applications/products	
Containers <i>Ongoing throughout project</i>	▪ Move newly built Linux containers to QA/Stage/Prod ▪ Integrate containers with legacy LMS	▪ Ongoing containerization and integration of LMS applications through QA/Staging/Prod	BAI DevOps owners identified, aligned and supportive of prioritized container activity Integrate at least one container with legacy LMS in production	Complete work started in FY23 and progress towards a new LMS architecture using containers and microservices to increase flexibility and performance	Most work performed by BAI developers; nvisia will focus on integration details

Summary

The following resources will be provided by nvisia. Work will be provided on a Time and Materials basis, starting on September 5, 2023 and completing on April 26, 2024. We approximate that we can conclude this project work within 8 months based on availability of consulting resources. nvisia and BAI will actively communicate and adjust timelines based on consultant availability in the coming months.

Resource Allocation

Resource	Hourly Rate	Allocation	Starting Date	Total
Delivery Lead Jason Militante	\$220	20%*	9/5/23	\$63,360
Client Partner Courtney Faulkner	\$240			

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Principal Architect TBD	\$215	100%	9/5/23	\$283,800
DevOps Lead Alex Searing	\$175	100%	9/5/23	\$231,000
Software Engineer TBD	\$155	100%	9/5/23	\$204,600
				\$782,760

*Delivery Lead and Client Partner will bill based on project needs, and combined will not exceed 20% allocation on project.

****BAI spend is to not exceed \$785,000.** BAI cost savings will largely depend on hiring a senior technical architect resource in time to own LMS microservice design work and/or SSO/Authentication planning and execution. (See Assumptions)

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Nvisia Roles and Responsibilities

Role	Responsibilities
Delivery Lead	✓ Scrum management and adherence to agile ceremonies ✓ Authors project plans and facilitates delivery expectations with BAI ✓ Track delivery tasks to plan ✓ Provide weekly status reporting and interface with BAI stakeholder group
Client Partner	✓ Provides technical direction and roadmap adhering to BAI business drivers/objectives ✓ Strategic guidance on approach and tooling based on deep industry experience and understanding of best practices ✓ Orchestrates project opportunities with nvisia capability areas and consultant skill sets
Principal Architect	✓ Lead technical architecture discussions and client enablement for solution ownership ✓ Point person for design implementation and task work ✓ Participate in code reviews and task planning discussions ✓ Interface with BAI technical leadership for alignment purposes
DevOps Lead	✓ Primary DevOps resource tasked with containerization of applications and platform configurations ✓ Supports architect direction and execution of DevOps related work ✓ Advises on cloud architecture and DevOps best practices
Software Engineer	✓ Adds additional velocity and support in completion of development tasks including POC development and testing

Costs

The estimated maximum cost is \$782,760. Work will be provided on a Time and Materials basis, where BAI will only be liable for actual hours worked and services/deliverables performed.

Invoicing will occur within the first 5 business days of each month for hours worked in the previous month and will be accompanied by detailed weekly timesheets. Payment terms are net 30 days from receipt of invoice.

Assumptions

This Statement of Work assumes the following:

- nvisia team will work under BAI guidance to determine product vision, feature priority, and implementation focus.
- BAI can allocate the equivalent of at least one senior software engineer 100% for the duration of this effort.
- BAI will hire a senior technical architect resource early in Q1 CY24 to own LMS microservice design architecture, SSO/Authentication plan, and LMS technical roadmap
- BAI will provide timely feedback regarding efforts and deliverables.
- nvisia will have access for collaboration and project delivery activities with BAI engineers, leads and testing teams.
- The nvisia team will work remotely. Core working hours and regular check-ins will be established as part of the Project Charter (Phase 1 deliverable). There may be occasional in-person meetings but they will be kept to a minimum.
- Per nvisia client relationship processes, a recurring sponsor-level touchpoint will be established.

Completion Criteria

nvisia shall have satisfied its responsibilities under this Statement of Work when 1) the Deliverables have been provided; or 2) nvisia has provided the specified number of person-days/hours of services; or 3) the mutually agreed upon project end date expires; or 4) BAI terminates this Statement of Work with a minimum of ten (10) business days' notice, whichever occurs first. In the event that BAI elects to terminate this SOW with less than the required ten (10) business days' notice, then for each business day less than the required ten (10) business days BAI has terminated this SOW (the "Early Termination Period"), client shall pay to nvisia as liquidated damages and not as a penalty, the amount of billings nvisia's consultants would have billed during the Early Termination Period, determined, if necessary, by the Estimated Cost Section of this SOW. For the avoidance of doubt, all invoicing by nvisia and payments by BAI shall operate in accordance with the procedures outlined in the Consulting Services Agreement (CSA).

Agreement Parameters

This SOW defines the scope of work to be accomplished by nvisia under the terms and conditions of the Consulting Services Agreement ("CSA"), signed between nvisia and BAI on May 1st, 2023. This CSA contains the general contract terms, which will govern the performance of the Services pursuant to the SOW. This CSA addresses payment terms, intellectual property rights, warranty, insurance, and other matters. This Statement of Work and the CSA (and any applicable Attachments) represent the complete agreement ("Agreement") between the parties regarding this transaction and replace any prior oral or written communications. Changes to the Statement of Work will be processed in accordance with the procedure described in the CSA.

The investigation and the implementation of changes may result in modifications to the Estimated Schedule, Charges, or other terms of this Agreement.



Under this SOW, nvisia will guide, assist and materially contribute to the Project by providing the Services and Deliverables as further set forth herein.

Commitment Signatures

At nvisia, **Your Success** is our success. On behalf of everyone at nvisia, thank you for the opportunity to present this statement of work to assist with your 2023 initiatives and goals. Should BAI choose to pursue this SOW, once executed, the formalized CSA and SOW will govern the relationship and supersede any prior communications.

Approval

By signing below, BAI and nvisia agree that the complete agreement between the two companies regarding these Services consists of 1) this SOW, 2) nvisia non-disclosure agreement (or any equivalent agreement signed by both of us) and 3) the CSA. Once signed, any reproduction of this SOW by reliable means (for example, photocopy or facsimile) is considered an original, unless prohibited by local law.

By signing below, BAI also agrees that this proposal will become the SOW effective for this project. No work shall begin under this SOW until an executed SOW is returned to nvisia.

If BAI internal processes require a purchase order in order for nvisia to invoice for the Services rendered under this SOW, the purchase order (which must reference this SOW number) should be faxed to nvisia's Chicago office at (312) 985-8101 before work commences.



Agreed to and Accepted by

nvisia

Shaun Lovick

Shaun Lovick (Oct 3, 2023 09:41 CDT)

Signature

Shaun Lovick

Name

Oct 3, 2023

Date

BAI

Julie Faber

Julie Faber (Oct 2, 2023 08:09 CDT)

Signature

Julie Faber

Name

Oct 2, 2023

Date

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BAI DevOps and LMS Microservice Arch SOW 09262023

Final Audit Report

2023-10-03

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